

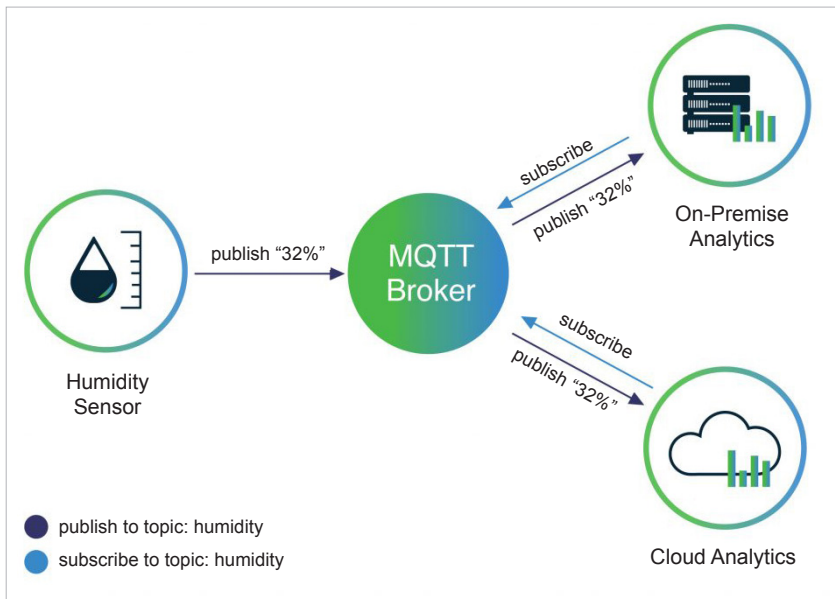


Figure 3: MQTT data transaction model (Courtesy: BehrTech)

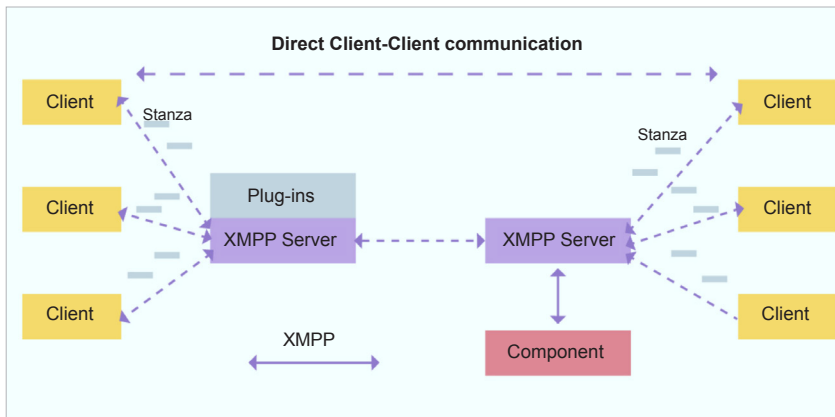


Figure 4: XMPP data transaction model (Courtesy: CometChat)

2 depicts the data transaction model of the CoAP protocol.

Message queuing telemetry transport (MQTT)

Perhaps the most widely used protocol in industry level IoT products, MQTT uses a publish/subscribe model for transacting data between different devices. In this model, devices are divided into three types, namely, the producers, consumers and brokers. Producers publish data under what are called topics. A topic is basically a channel under which related data can be published by the producer. The

consumer devices subscribed to that particular channel then receive the message. The entire transmission of data from a publisher to its subscribers is made possible with the help of brokers. Brokers receive messages from producers that they then filter into designated topics and share with the subscribers for each of the topics.

MQTT uses the TCP/IP protocol to enable communication between multiple devices. What makes it ideal for industry level products is that it simplifies sending and receiving messages between multiple devices. Figure 3 depicts the data transaction

model of MQTT with the help of an example of a smart humidity sensor. In the example, the humidifier is the producer that publishes data to the broker under separate topics. The consumers, which in this example are the on-premise analytics and the cloud, receive the data from the broker for their respective topics. So the producer publishes the data to the broker, and the broker publishes the data to the subscribers.

Extensible messaging and presence protocol (XMPP)

XMPP started as an instant messaging communication protocol but soon evolved into becoming an IoT protocol. It is an XML based server/client model protocol. The architecture, as depicted in Figure 4, is fairly straightforward. Clients connect to an XMPP server to communicate. The server facilitates the communication sessions between the clients. Since the medium of communication is in XML, it reduces the size of the data being transmitted and makes transactions faster.

There are, however, a few drawbacks in XMPP that do not make it an ideal solution for enterprise level IoT applications. For one, there is no quality-of-service feature in XMPP unlike CoAP, which makes it unreliable. Second, it doesn't have end-to-end encryption, which makes it insecure for sensitive data transmission.

Data distribution service (DDS)

DDS is similar to MQTT, but better. It follows a publish/subscribe model but does not require a broker for the transaction of data between devices. Add to that, it also supports multicasting and quality-of-service, which makes it ideal for usage in industry level products. The Object Management Group (OMG) first designed DDS with scalability in mind. Hence the brokerless model makes it highly scalable and adaptable to changing requirements.